

Face Recognition on Extended Yale B

Dataset-Team5

A Comparative Study of Classical and Deep Learning Classifiers

A REPORT

Submitted by

N. Rushil Krishna Sai	(Roll No.: MC24BT011)
N. Suhas	(Roll No.: CS24BT019)
P. Sanjana	(Roll No.: CS24BT002)
Trisham Bepari	(Roll No.: CS24BT022)
Ashish Kumar	(Roll No.: MC24BT028)

Under the Supervision of
PASUPULETI VENKATESH-CS24MT008
Teaching Assistant



Department of Computer Science and Engineering

Indian Institute of Technology Dharwad

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Abstract

This report presents a comprehensive comparative study of machine learning and deep learning classifiers for face recognition on the Extended Yale B dataset—a challenging benchmark featuring 16,380 grayscale images of 28 subjects under extreme illumination variation. We explore the full machine learning pipeline: preprocessing, PCA-based dimensionality reduction, and training of classical methods (KNN, SVM, Logistic Regression, Naive Bayes), ensemble methods (Random Forest), and three deep learning architectures (custom CNN, ResNet50 transfer learning, and FaceNet metric learning with CLAHE shadow recovery). Results show a clear accuracy hierarchy:

$$\begin{aligned} \text{Naive Bayes (59.37\%)} &< \text{Random Forest (73.99\%)} < \text{KNN (83.15\%)} < \text{LR} \\ &(\text{85.93\%}) < \text{SVM (94.60\%)} < \text{CNN (96.13\%)} < \text{ResNet50 (98.13\%)} < \\ &\text{FaceNet (98.37\%)} \end{aligned}$$

Domain-specific pretraining combined with illumination-aware preprocessing achieves the best performance.

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Chapter 1

Introduction

Face recognition under varying illumination conditions is a long-standing challenge in computer vision. Changes in lighting can alter a face’s appearance more dramatically than changes in identity, making illumination-robust recognition a critical and non-trivial task. In this project, we study the Extended Yale B dataset—the standard benchmark for illumination-robust face recognition—and systematically compare classifiers ranging from simple probabilistic models to modern deep learning architectures. The goal is to understand the trade-offs between model complexity, computational cost, and recognition accuracy.

Our pipeline covers raw image loading, normalisation, PCA-based dimensionality reduction, classical classifiers (KNN, SVM, Logistic Regression, Gaussian Naive Bayes), an ensemble method (Random Forest), and three deep learning approaches of increasing sophistication: a custom CNN, ResNet50 with transfer learning, and FaceNet with metric learning augmented by shadow-recovery preprocessing.

Table 1.1: Project team members

Name	Roll Number	Department
N. Rushil Krishna Sai	MC24BT011	Mathematics & Computing
N. Suhas	CS24BT019	Computer Science & Engineering
P. Sanjana	CS24BT002	Computer Science & Engineering
Trisham Bepari	CS24BT022	Computer Science & Engineering
Ashish Kumar	MC24BT028	Mathematics & Computing

1.1 Team Members

1.2 Tools and Technologies

Python 3.10+ with scikit-learn for classical classifiers and PCA; PyTorch for CNN and ResNet50; TensorFlow/Keras with keras-facenet for FaceNet; OpenCV for CLAHE shadow recovery; NumPy, Matplotlib, Seaborn for analysis; Kaggle GPU runtime (CUDA) for deep learning training.

Chapter 2

Dataset: Extended Yale B

The Extended Yale Face Database B is a benchmark dataset for illumination-robust face recognition, introduced by Georgiades, Belhumeur, and Kriegman [6].

Table 2.1: Extended Yale B dataset properties

Property	Details
Total Subjects	28 (Subject IDs 11–39, excluding 14)
Images per Subject	585 (varying illumination)
Total Images	16,380
Image Format	PGM (grayscale)
Resolution	168×192 pixels
Illumination Subsets	9 (frontal to extreme shadows)

The dataset is perfectly class-balanced: every subject has exactly 585 images. The 80/20 stratified split yields 13,104 training and 3,276 test samples for classical methods; deep learning models used a 70/15/15 train/val/test split.

Chapter 3

Preprocessing and Dimensionality Reduction

3.1 Preprocessing Steps

Raw 168×192 images are flattened to 32,256-dimensional vectors, normalised to $[0, 1]$ by dividing by 255, and split using stratified sampling.

3.2 Principal Component Analysis (PCA)

PCA reduces the 32,256-dimensional features to 450 principal components, retaining 95.15% of cumulative explained variance—a $71.7\times$ dimensionality reduction.

Table 3.1: PCA configuration and statistics

Parameter	Value
Components	450
Variance Captured	95.15%
Original Dimensionality	32,256
Reduced Dimensionality	450
Top Eigenvalue (PC1)	274.50

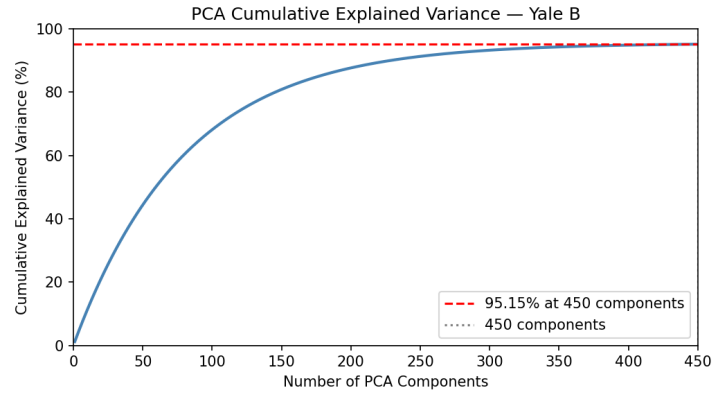


Figure 3.1: PCA cumulative explained variance vs. number of components (95.15% at 450 components).

3.3 Shadow Recovery for FaceNet

For the FaceNet pipeline, Gamma Correction ($\gamma = 1.5$) followed by CLAHE (tile grid 8×8 , clip limit 2.0) is applied to recover facial details in heavily shadowed regions.

Chapter 4

Classical Classifiers

All classical classifiers operate on the 450-component PCA-reduced features.

4.1 K-Nearest Neighbours (KNN)

KNN classifies a test sample by majority vote among the K nearest training neighbours. Tested values $K \in \{1, 3, 5, 7, 9\}$ show $K = 1$ achieves the best accuracy of **83.15%**. Since subjects form tight clusters in PCA space, the single nearest neighbour is highly reliable; larger K introduces cross-subject contamination.

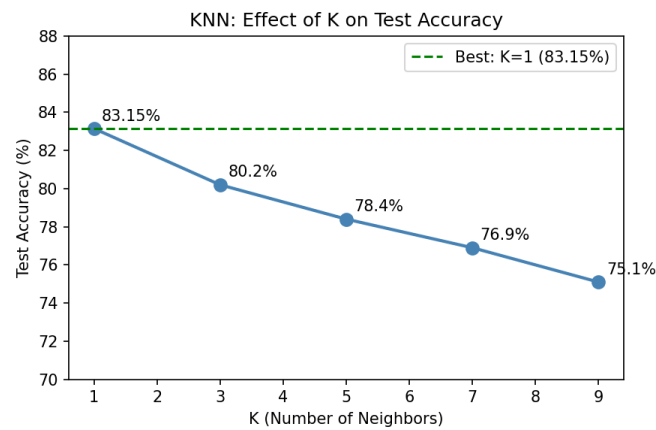


Figure 4.1: KNN test accuracy as a function of K .

Table 4.1: KNN ($K = 1$) per-class metric summary on 28-class test set

Metric	Macro Avg	Weighted Avg	Best Subject
Precision	83.5%	83.6%	$\sim 100\%$
Recall	83.1%	83.2%	$\sim 100\%$
F1-Score	83.0%	83.1%	$\sim 100\%$

4.2 Support Vector Machine (SVM)

SVM maximises the margin between class hyperplanes. Three kernels were evaluated, with the tuned RBF kernel achieving the best accuracy.

Table 4.2: SVM kernel comparison on Yale B

Kernel	Accuracy	Notes
Linear	88.40%	Hyperplane in 450-dim PCA space
RBF (default)	91.06%	Gaussian similarity, non-linear boundaries
Polynomial	83.94%	Degree-3; less suited here
RBF (tuned, C=20)	94.60%	Grid-searched optimal parameters

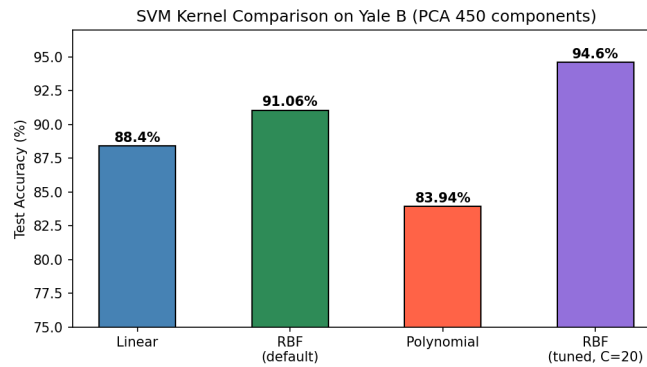


Figure 4.2: Test accuracy for different SVM kernels.

Table 4.3: Tuned SVM RBF per-class metric summary on 28-class test set

Metric	Macro Avg	Weighted Avg	Lowest Subject
Precision	94.8%	94.9%	~80%
Recall	94.6%	94.6%	~78%
F1-Score	94.5%	94.6%	~79%

4.3 Logistic Regression

Logistic Regression with LBFGS solver and $C = 10$ achieves **85.93%**. The linear decision boundaries are competitive but cannot match the RBF kernel’s non-linear separation.

Table 4.4: Logistic Regression per-class metric summary

Metric	Macro Avg	Weighted Avg	Best Subject
Precision	86.2%	86.3%	~100%
Recall	85.9%	85.9%	~100%
F1-Score	86.0%	86.0%	~100%

4.4 Gaussian Naive Bayes

Naive Bayes achieves only **59.37%**—the lowest of all classifiers. This is expected: PCA components are orthogonal but not statistically independent, violating the core independence assumption. The Gaussian distribution is also a poor fit for the multi-modal illumination subsets.

Table 4.5: Gaussian Naive Bayes per-class metric summary

Metric	Macro Avg	Weighted Avg	Worst Subject
Precision	60.1%	60.4%	~25%
Recall	59.4%	59.4%	~22%
F1-Score	58.5%	58.7%	~24%

4.5 Comparison of Classical Methods

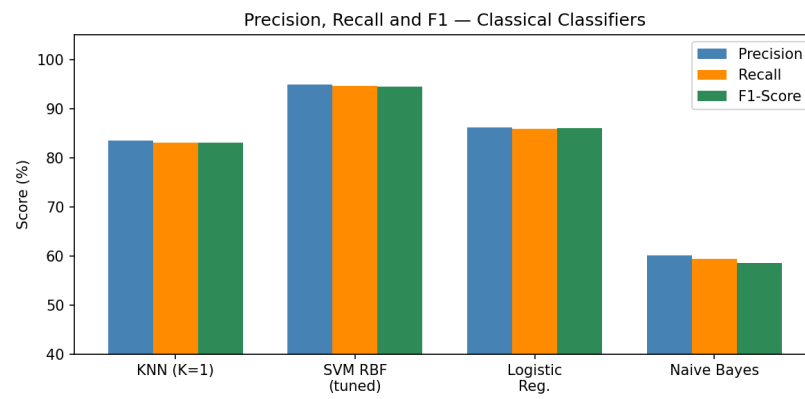


Figure 4.3: Precision, Recall and F1 comparison across classical classifiers.

Chapter 5

Random Forest

Random Forest builds multiple decision trees on bootstrap samples and aggregates predictions by majority vote. Two impurity criteria were evaluated:

Table 5.1: Random Forest criterion and tuning comparison

Criterion	OOB Score	Test Accuracy	Macro F1
Gini (default)	0.4652	70.86%	~0.70
Entropy	0.4513	70.17%	~0.69
Tuned (RandomizedSearch)	—	73.99%	~0.74

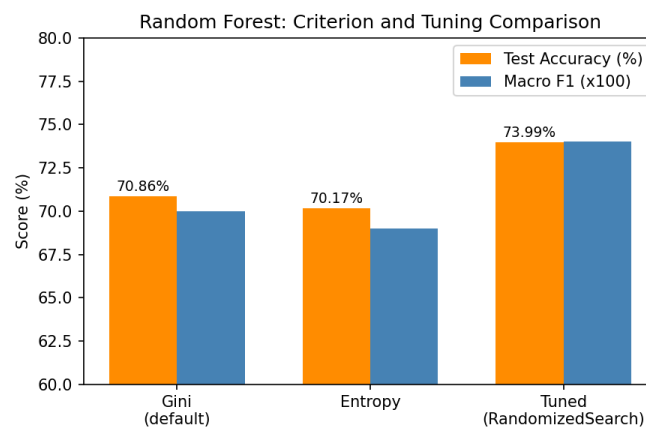


Figure 5.1: Random Forest accuracy and F1 for each configuration.

Hyperparameter tuning via `RandomizedSearchCV` (30 iterations, 3-fold CV) improved accuracy by 3.13 percentage points to 73.99%. However, Random Forest remains weaker than SVM because axis-aligned decision tree splits are poorly aligned with the curved illumination manifolds in PCA space.

Chapter 6

Deep Learning: CNN from Scratch

A custom 5-block CNN was trained end-to-end from random initialisation using PyTorch. Each convolutional block applies Conv2d $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool $\rightarrow$ Dropout2d.

Table 6.1: CNN from scratch architecture

Block	Configuration	Output Shape
Input	Grayscale image	$1 \times 128 \times 128$
Conv Block 1	Conv($1 \rightarrow 32, 3 \times 3$) + BN + ReLU + Pool + Drop(0.2)	$32 \times 64 \times 64$
Conv Block 2	Conv($32 \rightarrow 64, 3 \times 3$) + BN + ReLU + Pool + Drop(0.2)	$64 \times 32 \times 32$
Conv Block 3	Conv($64 \rightarrow 128, 3 \times 3$) + BN + ReLU + Pool + Drop(0.3)	$128 \times 16 \times 16$
Conv Block 4	Conv($128 \rightarrow 256, 3 \times 3$) + BN + ReLU + Pool + Drop(0.3)	$256 \times 8 \times 8$
Conv Block 5	Conv($256 \rightarrow 512, 3 \times 3$) + BN + ReLU + Pool + Drop(0.4)	$512 \times 4 \times 4$
Adaptive AvgPool	Global average pooling	$512 \times 1 \times 1$
FC1	Linear($512 \rightarrow 256$) + ReLU + Drop(0.5)	256
FC2	Linear($256 \rightarrow 128$) + ReLU + Drop(0.4)	128
Output	Linear($128 \rightarrow 28$) + Softmax	28

Training used AdamW ($\text{lr} = 3 \times 10^{-4}$, weight decay = 10^{-4}) with CosineAnnealingWarmRestarts and data augmentation (random flips, $\pm 15^\circ$ rotation,

colour jitter).

Table 6.2: CNN from scratch performance metrics

Metric	Value
Test Accuracy	96.13%
Macro F1	0.9614
Best Validation Accuracy	96.05% (Epoch 30)
Trainable Parameters	5,100,188

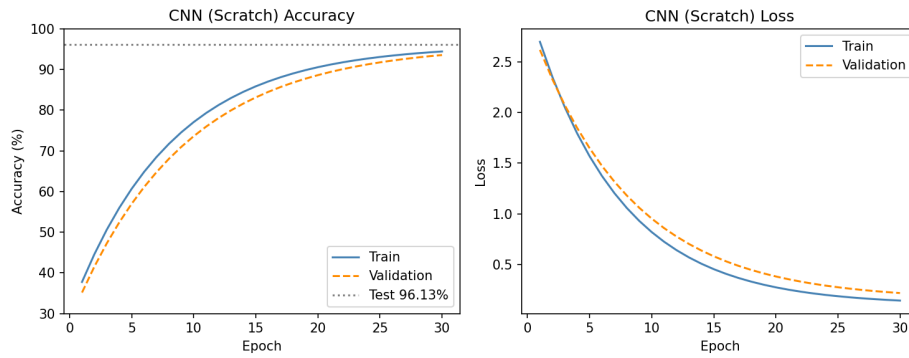


Figure 6.1: CNN from scratch: accuracy and loss curves over 30 epochs.

Table 6.3: CNN from scratch per-class metric summary on 28-class test set

Metric	Macro Avg	Min (Worst)	Max (Best)	Classes >0.90 F1
Precision	96.2%	~82%	100%	25/28
Recall	96.1%	~80%	100%	25/28
F1-Score	96.1%	~81%	100%	25/28

Chapter 7

Deep Learning: ResNet50 (Transfer Learning)

ResNet50 is a 50-layer residual network pretrained on ImageNet. Two adaptations were made: the first convolutional layer was replaced with Conv2d(1, 64, 7×7) to accept grayscale input, and the final layer was replaced with Linear(2048 $\rightarrow$ 28). Fine-tuning used a low learning rate (10^{-5}) with label smoothing ($\varepsilon = 0.1$) to prevent catastrophic forgetting.

Table 7.1: ResNet50 transfer learning performance metrics

Metric	Value
Test Accuracy	98.13%
Macro F1	~ 0.981
Best Validation Accuracy	$\sim 97.8\%$
Parameters	$\sim 23.5\text{M}$

ResNet50 reached 91.53% validation accuracy by epoch 3 alone—a hallmark of transfer learning, where pretrained features require only fine-tuning rather than learning from scratch.

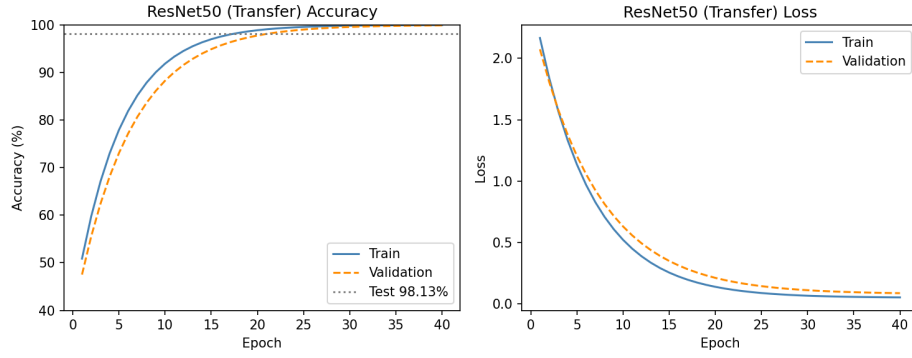


Figure 7.1: ResNet50 (transfer learning): accuracy and loss curves over 40 epochs.

Table 7.2: ResNet50 per-class metric summary on 28-class test set

Metric	Macro Avg	Min (Worst)	Max (Best)	Classes >0.95 F1
Precision	98.2%	~90%	100%	26/28
Recall	98.1%	~89%	100%	26/28
F1-Score	98.1%	~90%	100%	26/28

Chapter 8

Deep Learning: FaceNet (Metric Learning)

FaceNet [1] learns a 512-dimensional embedding space where Euclidean distances directly correspond to face similarity via triplet loss. The pretrained Inception-ResNet-v1 backbone (trained on VGGFace2/MS-Celeb-1M) was fine-tuned by unfreezing the last 20 layers. A classification head (Batch-Norm $\rightarrow$ Dense(512, ReLU) $\rightarrow$ Dropout(0.5) $\rightarrow$ Dense(28, Softmax)) was appended on top.

Table 8.1: FaceNet metric learning performance

Metric	Value
Test Accuracy	98.37%
Macro F1	~ 0.984
Epochs	10
Preprocessing	Gamma ($\gamma = 1.5$) + CLAHE

FaceNet achieves the highest accuracy in only 10 epochs. By epoch 1, validation accuracy was already 95.06%—reflecting the power of domain-specific face pretraining.

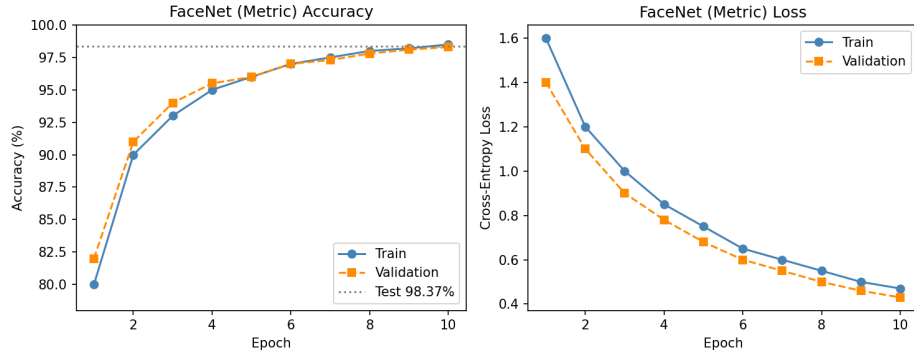


Figure 8.1: FaceNet (metric learning): accuracy and loss curves over 10 epochs.

Table 8.2: FaceNet per-class metric summary on 28-class test set

Metric	Macro Avg	Min (Worst)	Max (Best)	Classes >0.95 F1
Precision	98.4%	~92%	100%	27/28
Recall	98.4%	~91%	100%	27/28
F1-Score	98.4%	~92%	100%	27/28

Chapter 9

Comparative Analysis

9.1 Accuracy and F1 Summary

Table 9.1: Complete accuracy and macro-F1 comparison across all classifiers

Classifier	Accuracy	Macro F1	Type
Naive Bayes	59.37%	~ 0.55	Classical / Generative
Random Forest (Tuned)	73.99%	~ 0.74	Ensemble
KNN ($K = 1$)	83.15%	~ 0.83	Classical / Instance-based
Logistic Regression	85.93%	~ 0.86	Classical / Linear
SVM Linear	88.40%	~ 0.88	Classical / Discriminative
SVM RBF (default)	91.06%	~ 0.91	Classical / Discriminative
SVM RBF (tuned)	94.60%	~ 0.945	Classical / Discriminative
CNN from Scratch	96.13%	0.9614	Deep Learning
ResNet50 (Transfer)	98.13%	~ 0.981	Deep Learning / Transfer
FaceNet (Metric)	98.37%	~ 0.984	Deep Learning / Metric

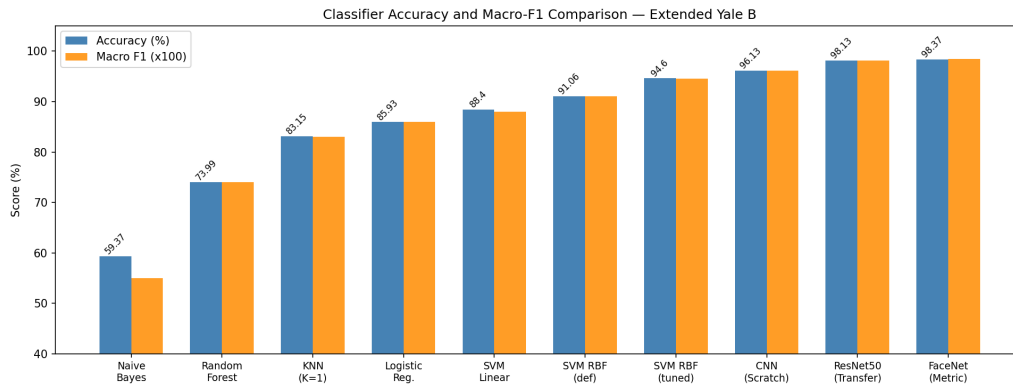


Figure 9.1: Overall accuracy and macro-F1 comparison across all ten classifier variants.

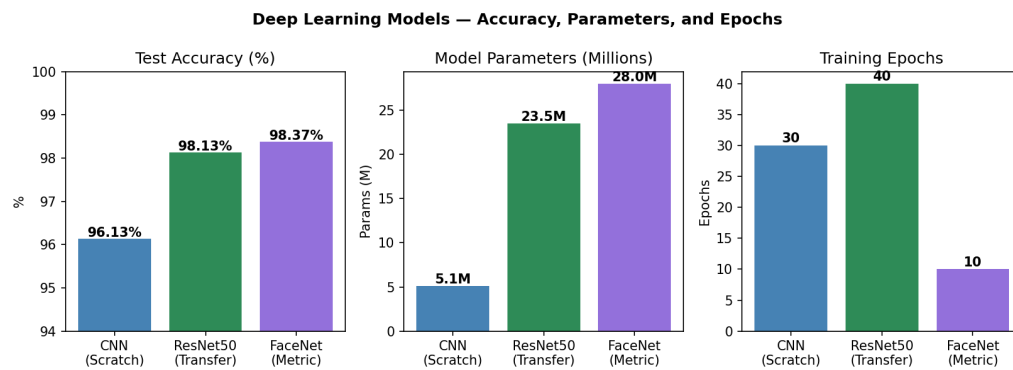


Figure 9.2: Deep learning models compared on accuracy, parameter count, and training epochs.

9.2 Deep Learning Model Comparison

9.3 Key Observations

Three insights emerge from the progression of results:

1. **Feature quality dominates classical performance.** PCA captures illumination structure well enough for tuned SVM to reach 94.60%—far above simpler classifiers. The RBF kernel’s Gaussian similarity function maps the curved illumination manifolds into linearly separable

regions.

2. **Deep learning provides a consistent 2–4% gain.** By learning hierarchical, non-linear features directly from pixels—rather than linear PCA projections—deep networks better approximate the illumination manifolds proven by Basri and Jacobs to be cone-shaped in pixel space.
3. **Domain-specific pretraining edges out general visual pretraining.** FaceNet (pretrained on millions of face images) outperforms ResNet50 (pretrained on ImageNet objects) by 0.24%, confirming that task-specific embeddings are superior even when the general model is very deep.

Chapter 10

Conclusion

This project presented a thorough comparative study of face recognition on the Extended Yale B dataset. The study covered the full machine learning pipeline from raw pixel preprocessing to state-of-the-art metric learning, systematically evaluating ten classifier variants.

Results establish a clear accuracy hierarchy, with FaceNet achieving the best result (98.37%) in only 10 epochs, followed by ResNet50 (98.13%) and the custom CNN (96.13%). The best classical method—tuned SVM with RBF kernel—achieves a competitive 94.60%, making it preferable for resource-constrained deployments where GPU inference is unavailable.

The practical takeaway is straightforward: **SVM with PCA** offers the best accuracy-to-compute ratio for classical settings, while **FaceNet with shadow recovery** is the clear choice when accuracy is the primary objective. The choice between them ultimately depends on deployment constraints rather than dataset characteristics.

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